Computational Work 2: Cholesky Factorization

Éderson Passos Silva Registration: 00318375

Course: Computational Matrix Methods

Professor: Cibele Aparecida Ladeia

Abstract

This work investigates two least-squares problems using data derived from the student's enrollment number. First, an overdetermined linear system $A\mathbf{x} \approx \mathbf{b}$, with $A \in \mathbb{R}^{5\times4}$, is solved using the QR factorization $A = QR$. The least-squares solution is obtained by solving the resulting upper triangular system. The existence of a unique solution is discussed in terms of the rank of the coefficient matrix.

Second, a quadratic polynomial is fitted to six experimental data points using the least-squares formulation. The associated design matrix is constructed, and the polynomial coefficients are determined computationally. The quality of the fit is assessed by plotting the experimental data and the fitted curve and calculating the mean squared error.

The numerical implementation explicitly constructs the principal linear algebra algorithms, avoiding high-level routines for QR factorization and least-squares solutions. This approach connects the mathematical foundations of least squares with their computational implementation and numerical evaluation.

Keywords: QR factorization; least squares; quadratic polynomial fitting; numerical linear algebra.

1 Problem

The computational work is divided into two problems involving numerical linear algebra and the numerical solution of a steady-state heat conduction problem.

1.1 Problem 1: Symmetric Positive Definite Matrix

Let $A \in \mathbb{R}^{10\times10}$ and $b \in \mathbb{R}^{10}$ be generated according to the procedure given in Code 1 of Appendix A.1, using the student's enrollment number as the random seed.

The following tasks are considered:

1. Determine whether the matrix A is symmetric positive definite. If so, compute its Cholesky factorization
$$A = GG^T,$$
and present the Cholesky factor G.

2. Compute the condition number $\kappa(A)$ and justify the result.

3. Solve the linear system
$$Ax = b.$$

1.2 Problem 2: One-Dimensional Steady-State Heat Conduction

Consider the one-dimensional steady-state heat conduction problem in a unit-length bar,

$$-k\frac{d^2T}{dx^2} = f, \qquad 0 \le x \le 1, \tag{1}$$

where $T(x)$ denotes the temperature, $k > 0$ is the thermal conductivity, and f is a constant source term.

The boundary conditions are determined according to the procedure given in Code 2 of Appendix A.2. One of the following cases is selected:

$$\begin{aligned}
&\text{Case 1:} \quad \left.\frac{dT}{dx}\right|_{x=0} = 0, \qquad T(1) = 0,\\
&\text{Case 2:} \quad T(0) = 0, \qquad T(1) = 0,\\
&\text{Case 3:} \quad \left.\frac{dT}{dx}\right|_{x=0} = 0, \qquad \left.\frac{dT}{dx}\right|_{x=1} = 0.
\end{aligned}$$

Using the finite difference formulation specified in the assignment, with 11 grid points, the problem is represented by the linear system

$$MT = s, \tag{2}$$

where M is the matrix associated with the discretized differential operator, s is the source vector including the boundary-condition contributions, and T is the vector of unknown temperatures.

The objectives are:

1. Generate M and s using the student's enrollment number and solve the resulting linear system.
2. Present the numerical temperature distribution $T(x)$.
3. Plot the temperature distribution and interpret the result from a physical perspective.

2 Methodology

The computational experiments were performed using Python and the NumPy library. All problem-specific data were generated deterministically from the student's enrollment number, following the algorithms provided in the assignment.

2.1 Matrix Generation and Cholesky Factorization

For the first problem, the matrix $A \in \mathbb{R}^{10\times 10}$ is generated according to

$$A = BB^T + 10I, \tag{3}$$

where B is a randomly generated integer matrix and I is the 10×10 identity matrix. The vector $b \in \mathbb{R}^{10}$ is independently generated using the prescribed random seed.

The symmetry and positive definiteness of A are verified before computing its Cholesky factorization. When A is symmetric positive definite, the factorization

$$A = GG^T \tag{4}$$

is computed, where G is a lower triangular matrix with positive diagonal entries.

The condition number of A is then computed to quantify the sensitivity of the corresponding linear system to perturbations [1]. Finally, the system

$$Ax = b \tag{5}$$

is solved numerically.

2.2 Heat Conduction Problem

For the second problem, the parameters k, f, and the boundary-condition type are generated using the prescribed procedure and the enrollment number as the random seed. The spatial domain $[0, 1]$ is represented by 11 equally spaced grid points, resulting in a grid spacing

$$h = \frac{1}{10}. \tag{6}$$

The matrix M and vector s are constructed according to the finite difference formulation specified in the assignment. The resulting linear system

$$MT = s \tag{7}$$

is then solved numerically to obtain the temperature vector

$$T = \begin{bmatrix} T_0 & T_1 & \cdots & T_{10} \end{bmatrix}^T. \tag{8}$$

The numerical temperature values are associated with the corresponding spatial grid points and plotted as a function of x. The resulting temperature distribution is subsequently interpreted in terms of the physical behavior of the one-dimensional heat conduction problem.

All numerical computations were performed using double-precision floating-point arithmetic. For convergence analysis, the discrete L_2 error is explicitly defined as

$$\|e\|_{2,h} = \sqrt{h}\,\|e\|_2, \tag{9}$$

which is the exact formulation implemented in the computational experiment.

3 Results and Discussion

3.1 Problem 1: Cholesky Factorization and Linear System

The prescribed matrix $A \in \mathbb{R}^{10\times 10}$ is symmetric by construction. Furthermore, it is strictly positive definite. The algorithmic assembly defines $A = BB^T + 10I$. For any non-zero vector $x \in \mathbb{R}^{10}$, the quadratic form is bounded strictly away from zero:

$$x^T Ax = x^T(BB^T + 10I)x = \|B^T x\|_2^2 + 10\|x\|_2^2 > 0. \tag{10}$$

Consequently, A is symmetric positive definite (SPD), guaranteeing the existence and uniqueness of the Cholesky factorization $A = GG^T$.

The numerical computation of the lower triangular Cholesky factor G yields:

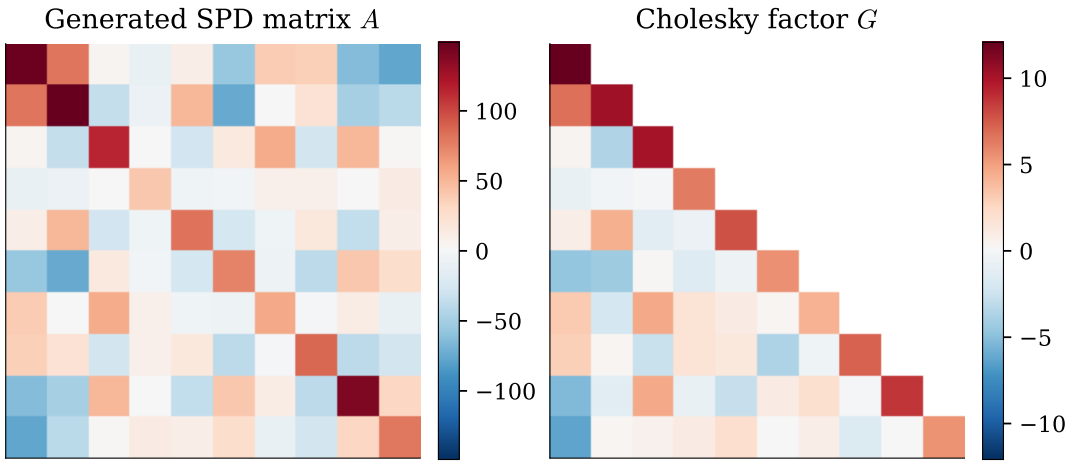


Figure 1: Structure of the generated symmetric positive-definite matrix A and its lower-triangular Cholesky factor G.

$$G = \begin{pmatrix}
10.9545 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0.5477 & 11.0770 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
2.1909 & -2.5458 & 8.4687 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
-5.6598 & 4.7937 & 4.6765 & 6.7169 & 0 & 0 & 0 & 0 & 0 & 0 \\
-1.2780 & -7.6104 & -1.0125 & -1.9379 & 6.9763 & 0 & 0 & 0 & 0 & 0 \\
3.0125 & -1.1420 & -4.7832 & -0.9092 & -1.3540 & 6.4095 & 0 & 0 & 0 & 0 \\
-6.2988 & -0.5010 & 3.0140 & -0.1999 & -5.6189 & 0.3166 & 8.2023 & 0 & 0 & 0 \\
-2.3735 & 1.9229 & -5.0663 & -1.6315 & -2.2491 & 1.3393 & -1.3539 & 9.0363 & 0 & 0 \\
0.8216 & 2.0357 & 2.7611 & -0.3009 & 1.3984 & 0.7296 & -1.8971 & 2.8919 & 7.3496 & 0 \\
-2.5560 & -3.4847 & -3.3383 & -1.3623 & -0.6890 & -1.7815 & 3.1255 & -0.9333 & 0.5297 & 5.7231
\end{pmatrix} \tag{11}$$

To verify the factorization, the reconstruction residual was calculated as $\|A - GG^T\|_2 \approx 2.001 \times 10^{-14}$, confirming reconstruction to near machine precision.

Since A is SPD, the 2-norm condition number can be computed as $\kappa_2(A) = \lambda_{\max}(A)/\lambda_{\min}(A)$. Using the actual computed matrix, the extreme eigenvalues are approximately $\lambda_{\min} = 10.1083$ and $\lambda_{\max} = 387.5890$, and therefore $\kappa_2(A) = 38.3435$ [1]. Using the factor G, the linear system $Ax = b$

was solved via forward and backward substitution. The computed solution vector is:

$$x = \begin{pmatrix} -0.2703 & 0.3296 & 0.1891 & 0.6653 & 0.1022 & 0.5456 & -0.1468 & -0.1706 & -0.1793 & -0.6184 \end{pmatrix}^T. \tag{12}$$

The algebraic accuracy of this solution is verified by the residual norm $\|Ax - b\|_2 \approx 9.0686 \times 10^{-15}$.

3.2 Problem 2: Original Finite-Difference Formulation

The original discrete problem was executed exactly as prescribed by Code 2 using the seed 318375. This generated the parameters $k = 0.0847$, $f = 0.6000$, grid spacing $h = 0.1$, and selected the Dirichlet–Dirichlet boundary condition corresponding to $T(0) = 0$ and $T(1) = 0$.

The resulting linear system $M_{orig}T_{orig} = s_{orig}$ was solved computationally. The numerical residual is $\|M_{orig}T_{orig} - s_{orig}\|_2 \approx 2.9899 \times 10^{-16}$. The computed solution satisfies the generated discrete linear system to near machine precision. However, this algebraic residual is fundamentally distinct from and does not imply consistency with the stated continuous PDE.

The assignment mathematically defines the steady-state heat equation as:

$$-kT''(x) = f, \quad x \in (0, 1). \tag{13}$$

The physical configuration of this boundary-value problem is illustrated in Figure 2.

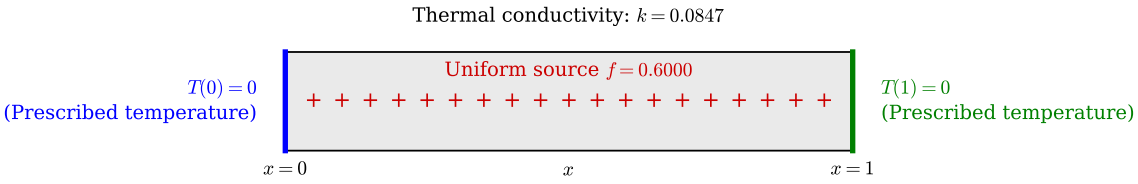


Figure 2: One-dimensional steady-state heat-conduction configuration for the generated Dirichlet–Dirichlet boundary conditions.

Subject to the Dirichlet–Dirichlet boundary conditions, the exact analytical solution can be derived by integrating the constant source term twice:

$$T_{exact}(x) = \frac{f}{2k}x(1 - x). \tag{14}$$

Given $k, f > 0$, the analytical solution has specific qualitative properties: it is non-negative on $[0, 1]$, strictly concave downward ($T''(x) = -f/k < 0$), reaches a global maximum at the center of the domain $x = 0.5$ ($T_{exact}(0.5) \approx 0.8855$), and decreases to the fixed temperature at both boundaries.

A rigorous diagnosis reveals that the prescribed discrete formulation is not consistent with the continuous differential equation stated in the assignment. Specifically:

1. **Sign inconsistency:** The interior matrix coefficients $(1/h^2, -2/h^2, 1/h^2)$ approximate the positive second derivative T_i'', omitting the required negative sign in the Laplacian operator.

2. **Source scaling inconsistency:** The generated right-hand side is $s_i = (h^2/k)f$, which improperly scales the source by h^2. Together with the sign inconsistency, the interior nodes inadvertently model the equation $T'' = \frac{h^2}{k}f$.

Consequently, at the prescribed grid resolution $h = 0.1$, the original formulation produces a qualitatively different temperature profile. As shown in Table 1, its maximum absolute error relative to the analytical solution of the stated continuous problem is approximately 0.8944. This discrepancy is attributable to the inconsistency of the discrete equations with the stated continuous model, rather than to failure of the linear solver.

3.3 Problem 2: Corrected Formulation

To construct a finite-difference system mathematically consistent with $-kT'' = f$, a corrected formulation was implemented. For the interior nodes ($1 \leq i \leq 9$), the standard central difference approximation was used:

$$-\frac{1}{h^2}T_{i-1} + \frac{2}{h^2}T_i - \frac{1}{h^2}T_{i+1} = \frac{f}{k}. \tag{15}$$

For the Dirichlet boundaries at $x = 0$ and $x = 1$, they are strictly enforced via:

$$T_0 = 0 \quad \text{and} \quad T_{10} = 0. \tag{16}$$

This yields a system that is an exact discrete representation of the continuum equation for quadratic solutions. This corrected linear system, denoted $M_{corr}T_{corr} = s_{corr}$, was solved yielding the corrected numerical solution shown in Table 1.

Table 1: Comparison of temperature solutions at $h = 0.1$. The original solution results from a literal execution of the prescribed code. The corrected solution implements the consistent formulation.

x	T_{exact}	T_{orig}	T_{corr}	$\lvert T_{orig} - T_{exact}\rvert$	$\lvert T_{corr} - T_{exact}\rvert$
0.0	0.0000	0.0000	-0.0000	5.7980e-18	1.1564e-15
0.1	0.3188	-0.0032	0.3188	3.2198e-01	9.9920e-16
0.2	0.5667	-0.0057	0.5667	5.7241e-01	8.8818e-16
0.3	0.7439	-0.0074	0.7439	7.5129e-01	7.7716e-16
0.4	0.8501	-0.0085	0.8501	8.5862e-01	5.5511e-16
0.5	0.8855	-0.0089	0.8855	8.9440e-01	4.4409e-16
0.6	0.8501	-0.0085	0.8501	8.5862e-01	4.4409e-16
0.7	0.7439	-0.0074	0.7439	7.5129e-01	2.2204e-16
0.8	0.5667	-0.0057	0.5667	5.7241e-01	1.1102e-16
0.9	0.3188	-0.0032	0.3188	3.2198e-01	0.0000e+00
1.0	0.0000	0.0000	0.0000	0.0000e+00	0.0000e+00

In contrast to the original formulation, the corrected numerical solution reaches $T_{corr}(0.5) \approx 0.8855$. Its maximum absolute error relative to the exact analytical solution drops to approximately 1.1564×10^{-15}. In exact arithmetic, the corrected centered finite-difference discretization reproduces the quadratic analytical solution exactly at the grid points; the observed discrepancies are attributable to floating-point roundoff.

3.4 Comparison and Discretization Accuracy

Figure 3 visually emphasizes the qualitative distinction between the formulations. The corrected formulation reproduces the qualitative features of the analytical solution precisely at all grid points. The interior stencil of the original formulation approximates T'' with the opposite sign and the source term is improperly scaled by h^2.

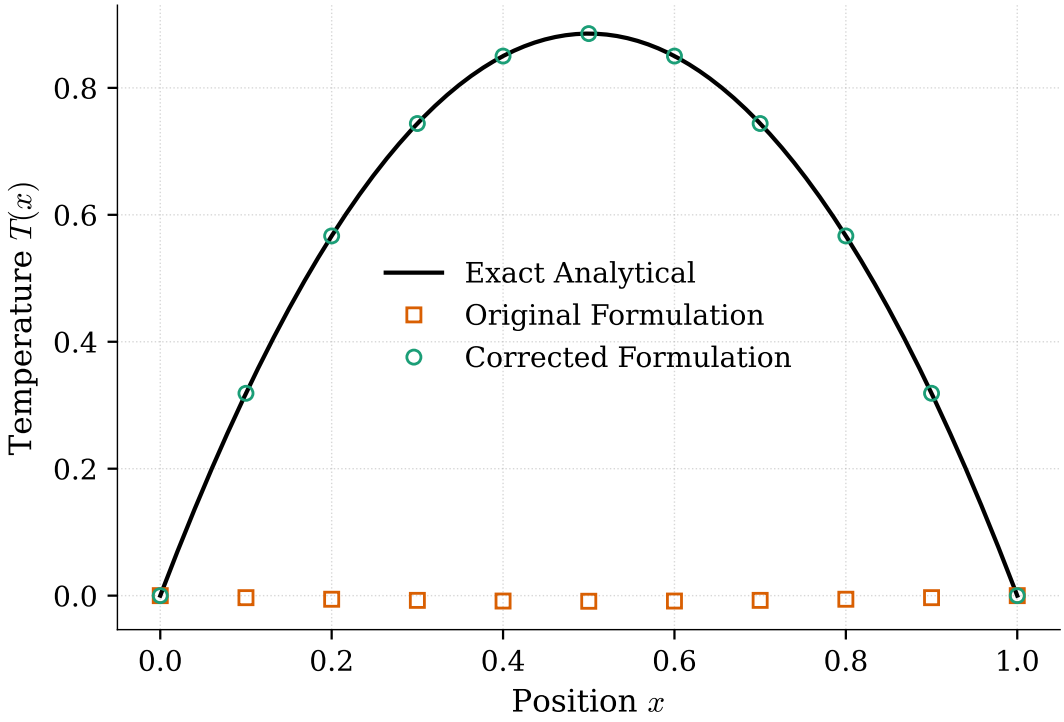


Figure 3: Comparison of the analytical solution with the profiles obtained from the original and corrected finite-difference formulations.

To validate the corrected method, an auxiliary convergence experiment was performed using grid sizes $N \in \{11, 21, 41, 81\}$. As shown in Table 2, the error remains at the level of floating-point roundoff across all resolutions.

Table 2: Empirical verification of the corrected finite-difference formulation. Errors represent floating-point-level discrepancies; no meaningful empirical convergence rate can be inferred.

N	h	L_∞ Error	Discrete L_2 Error
11	0.1000	1.1564e-15	6.7028e-16
21	0.0500	2.3315e-15	1.6846e-15
41	0.0250	2.6689e-14	1.6860e-14
81	0.0125	1.0254e-13	5.5344e-14

Because the exact solution is quadratic, the centered finite-difference discretization is exact in exact arithmetic; the reported discrepancies therefore reflect floating-point roundoff rather than a measurable asymptotic convergence regime.

3.5 Physical Interpretation

The stated steady-state boundary-value problem describes a one-dimensional bar subject to a constant internal heat source $f > 0$. The homogeneous Dirichlet boundary conditions at $x = 0$ and $x = 1$ fix the edges to a baseline reference temperature of zero. Physically, thermal energy is continuously generated inside the domain and must flow outwards through both boundaries to dissipate. This mechanism produces a symmetric temperature profile that is concave downward and reaches its maximum exactly at the midpoint of the domain. The corrected numerical solution reproduces these qualitative features perfectly, matching the analytical profile at all grid points.

By contrast, the original algorithmic formulation uses a discrete operator with the sign of the positive second derivative, together with the nonstandard source scaling described above. At the prescribed grid, this produces a concave-up profile with temperatures below the reference value imposed at the boundaries. The negative profile is inconsistent with the analytical solution of the stated PDE under the specified parameters and boundary conditions. This behavior is a consequence of the inconsistency between the generated discrete system and the stated differential equation, not evidence of numerical instability in the linear solve.

References

[1] Golub, G. H., & Van Loan, C. F. (2013). *Matrix Computations* (4th ed.). Johns Hopkins University Press.

[2] LeVeque, R. J. (2007). *Finite Difference Methods for Ordinary and Partial Differential Equations: Steady-State and Time-Dependent Problems*. SIAM.

A Generation Codes

The following codes reproduce the procedures provided in the assignment for generating the numerical data used in the computational experiments.

A.1 Code 1: Matrix and Vector Generation

Listing 1: Generation of the matrix A and vector b using the student's enrollment number.

```
import numpy as np

def gerar_matriz(matricula, n=10):
    np.random.seed(matricula)
    B = np.random.randint(-5, 6, size=(n, n))
    A = B @ B.T + n * np.eye(n)
    return A

def gerar_vetor(matricula, n=10):
    np.random.seed(matricula + 1)
    return np.random.randint(-20, 21, size=n)

matricula = 318375  # student's enrollment number

A = gerar_matriz(matricula)
b = gerar_vetor(matricula)

print("Matriz␣A:")
print(A)

print("\nVetor␣b:")
print(b)
```

A.2 Code 2: Heat Conduction Problem

Listing 2: Generation of the finite difference system for the one-dimensional heat conduction problem.

```
import numpy as np

def gerar_matriz_por_matricula(matricula, n=11):
    np.random.seed(matricula)

    h = 1 / (n - 1)
    M = np.zeros((n, n))

    k = np.random.uniform(0.05, 0.3)
    fonte = np.random.uniform(0.5, 2.0)

    tipo_bc = np.random.choice([
        "neumann-dirichlet",
```

```
        "dirichlet-dirichlet",
        "neumann-neumann"
    ])

    coef = 1 / h**2
    s = np.ones(n) * (h**2 / k) * fonte

    for i in range(1, n - 1):
        M[i, i - 1] = coef
        M[i, i] = -2 * coef
        M[i, i + 1] = coef

    if tipo_bc == "neumann-dirichlet":
        M[0, 0] = -coef
        M[0, 1] = coef
        M[-1, -1] = 1
        s[-1] = 0

    elif tipo_bc == "dirichlet-dirichlet":
        M[0, 0] = 1
        M[-1, -1] = 1
        s[0] = 0
        s[-1] = 0

    elif tipo_bc == "neumann-neumann":
        M[0, 0] = -coef
        M[0, 1] = coef
        M[-1, -2] = -coef
        M[-1, -1] = coef

    return M, s, k, fonte, tipo_bc


matricula = 318375  # student's enrollment number

M, s, k, fonte, bc = gerar_matriz_por_matricula(matricula)

print("Matriz␣M:\n", M)
print("Vetor␣s:\n", s)
print("k␣=", k)
print("fonte␣=", fonte)
print("BC␣=", bc)
```